

7.1 Point Estimation

When the random variable X follows a certain distribution with **unknown** $E(X) = \mu$ and **unknown** $V(X) = \sigma^2$, then statements about the size of μ and σ^2 , such as:

$$\mu \approx 12$$

$$\mu \geq 10$$

$$2 < \sigma^2 < 5$$

can be verified, if we have measurements of X , i.e. if we have a **random sample** of X .

Suppose $X_1, X_2, \dots, X_n$ is a **random sample** of certain random variable X with parameters $E(X) = \mu$ and $V(X) = \sigma^2$. So $X_1, X_2, \dots, X_n$ are mutually independent and they all have the same probability distribution.

The quantities based on $X_1, X_2, \dots, X_n$:

$$X_{\text{sum}} = X_1 + X_2 + \dots + X_n$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{sample mean}$$

$$\begin{aligned} S^2 &= \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \\ &= \frac{1}{n-1} \{ \sum_{i=1}^n X_i^2 - n \cdot \bar{X}^2 \} \quad \text{sample variance} \end{aligned}$$

are called statistics.

In Chapter 5, we have already established:

$$E(X_{\text{sum}}) = n \cdot \mu; \quad V(X_{\text{sum}}) = n \cdot \sigma^2$$

$$E(\bar{X}) = \mu; \quad V(\bar{X}) = \frac{\sigma^2}{n}$$

Statistics are used to make statements about the size of all kinds of parameters, such as μ and σ^2 .

In that case, it is also useful to know that both X_{sum} and $\bar{X}$ are approximately **normally distributed**, provided that n is large enough. This result is known as the Central Limit Theorem.

7.2 Sampling Distributions and the Central Limit Theorem

When the random variable X_{sum} is based upon a **random sample** of some random variable X , then, for large values of n , X_{sum} appears to follow a **normal distribution**. This property is known as the “**central limit theorem**” and is even visible for small values of n .

Example 1

Tossing n times with a **fair** die. Let X_i = number of dots at the i^{th} turn ($i = 1, 2, \dots, n$). Then X_{sum} is viewed.

- $n = 1$, i.e. $X_{\text{sum}} = X_1$

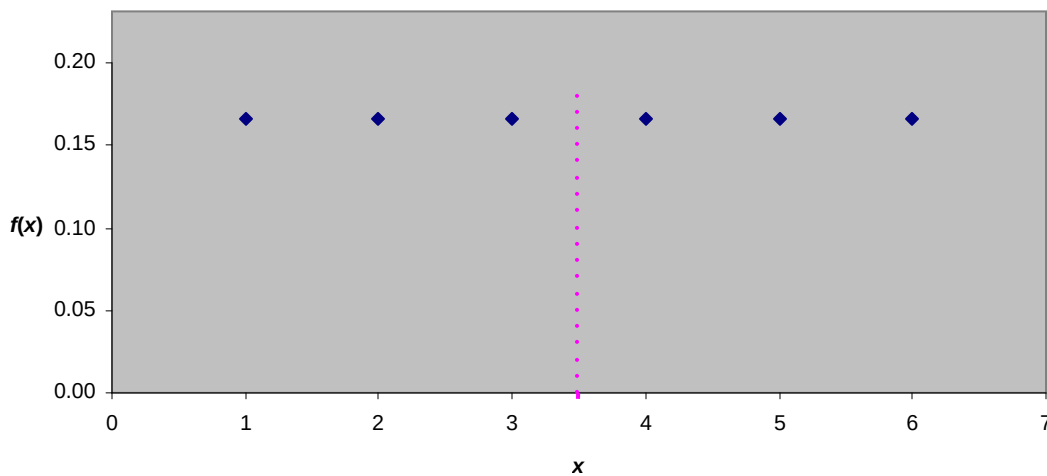


Figure 7.1 Graph of $f(x) = P(X_{\text{sum}} = x)$

- $n = 2$, i.e. $X_{\text{sum}} = X_1 + X_2$

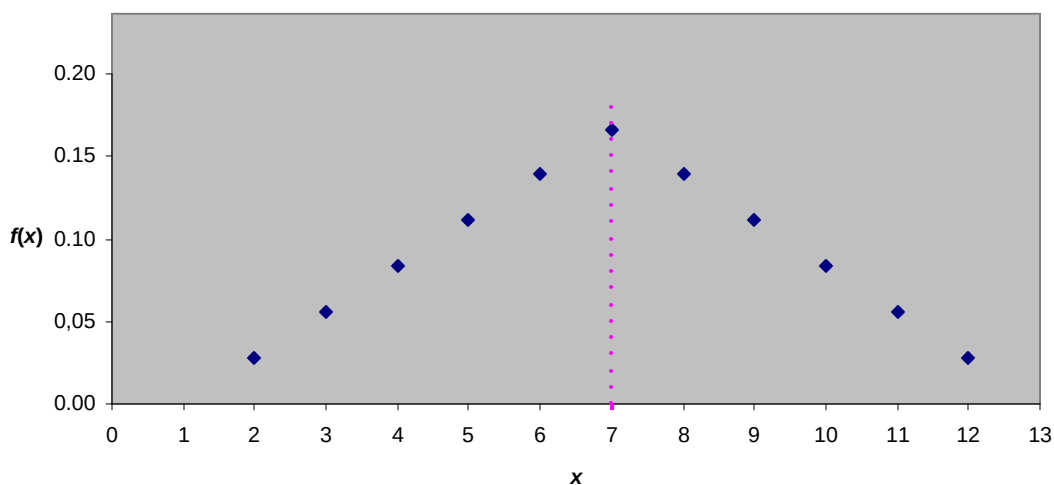


Figure 7.2 Graph of $f(x) = P(X_{\text{sum}} = x)$

- $n = 3$, i.e. $X_{\text{sum}} = X_1 + X_2 + X_3$

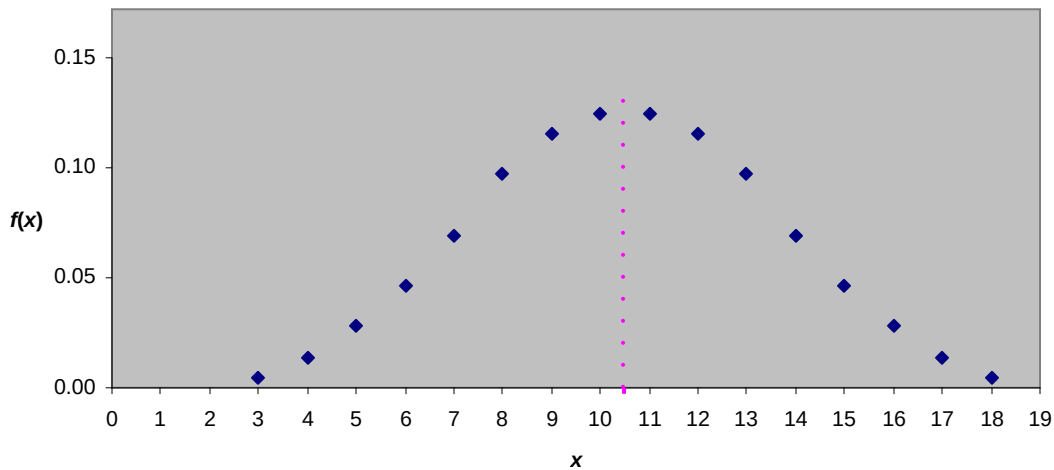


Figure 7.3 Graph of $f(x) = P(X_{\text{sum}} = x)$

Summarizing

Let $X_1, X_2, \dots, X_n$ be a **random sample** of certain random variable X with parameters $E(X) = \mu$ and $V(X) = \sigma^2$. Then, for large values of n :

$$X_{\text{sum}} = X_1 + X_2 + \dots + X_n \approx N(\mu_{\text{sum}} = n \cdot \mu; \sigma_{\text{sum}} = \sqrt{n} \cdot \sigma)$$

Example 2

Someone randomly chooses 25 times a *real* number between 0 and 4. What is the probability that the sum of these 25 numbers exceeds 70?

Solution

$$X_{\text{sum}} = X_1 + X_2 + \dots + X_{25}, \quad \text{with} \quad X_i \sim u[0; 4] \quad \text{continuous}$$

$$\mu = \frac{1}{2}(a + b) = 2$$

$$\sigma^2 = \frac{1}{12}(b - a)^2 = 4/3$$

According to the **Central Limit Theorem**:

$$X_{\text{sum}} \approx N(\mu_{\text{sum}} = 25 \cdot 2 = 50; \sigma_{\text{sum}} = \sqrt{25} \cdot \sqrt{\frac{4}{3}} = 5.7735)$$

So

$$P(X_{\text{sum}} > 70) \approx \text{normalcdf}(70, 10^{10}, 50, 5.7735)$$

$$= 0.0003$$

Remarks

- The **Central Limit Theorem** also applies to the sample mean:

$$\bar{X} \approx N(\mu_{\text{gem}} = \mu; \sigma_{\text{gem}} = \frac{\sigma}{\sqrt{n}})$$

- In the special case that X is **normally distributed**, “ $\sim$ ” applies instead of “ $\approx$ ”, regardless of the size of n .
- In the case that X is a **discrete** random variable, the normal approximation does not always lead to an accurate answer. We therefore apply a **correction for continuity**.

Example 3

Someone randomly chooses 25 times an *integer* from $\{0, 1, 2, 3, 4\}$. What is the probability that the sum of these numbers exceeds 70?

Solution

$$X_{\text{sum}} = X_1 + X_2 + \dots + X_{25},$$

with

$$X_i \sim u[0; 4]$$

discrete

$$\mu = \frac{1}{2}(a + b) = 2$$

$$\sigma^2 = \frac{(b-a+1)^2 - 1}{12} = 2$$

According to the **Central Limit Theorem**:

$$X_{\text{sum}} \approx N(\mu_{\text{sum}} = 25 \cdot 2 = 50; \sigma_{\text{sum}} = \sqrt{25} \cdot \sqrt{2} = 7.0711)$$

So

$$\begin{aligned} P(X_{\text{sum}} > 70) &\approx \text{normalcdf}(70.5, 10^{10}, 50, 7.0711) \\ &= 0.0019 \end{aligned}$$

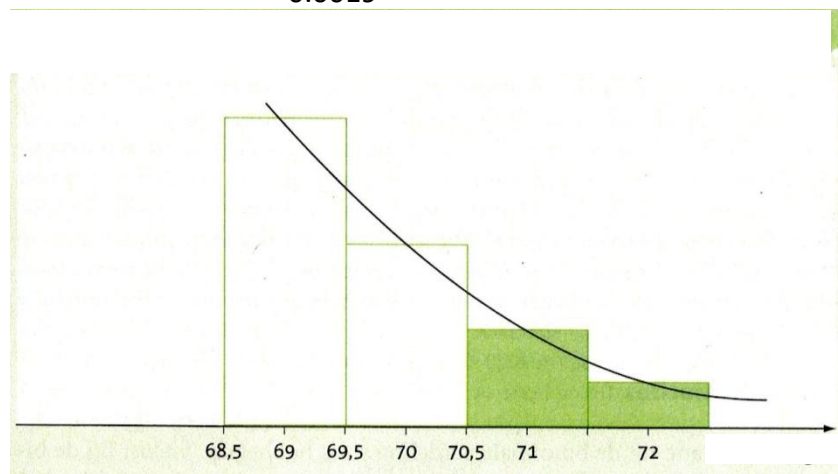


Figure 7.4

Correction for continuity